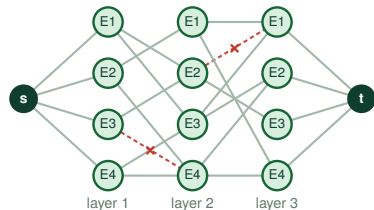


Confluence: routing Mixture-of-Experts with a conserved flow

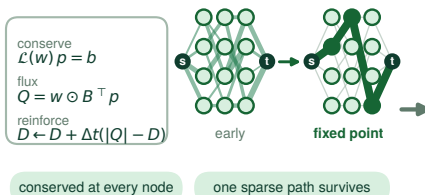
One legal path of experts per input, trained end to end, and a condition for when coupled routing beats an independent gate

A Per-input legal expert graph



Edges are exactly the legal expert transitions.
An illegal transition (x) has no edge, so it can never be used.

B Conserved-flow dynamic (Physarum)



The encoder sets per-edge costs; the flow relaxes to a single legal path of experts (no learned gate).

C Route and compose

Conserved flow (ours)



one legal path, 1 expert/layer: $\hat{y} = f_{E4} \circ f_{E1} \circ f_{E2}(x)$

Top-k gate (standard MoE)



The gate scores each layer independently and can chain experts in an order the task forbids.

D Train end to end: implicit differentiation

fixed point
 $D^* = F(D^*, \theta)$
backward solve
 $(I - \partial_D F)^T u = g$
gradient
 $\nabla_\theta = (\partial_\theta F)^T u$

$O(E)$ memory, any depth

matrix-free VJP + CG

checked vs. finite diff.

Gradients flow through the solver without unrolling it, so training cost stays linear in edges.

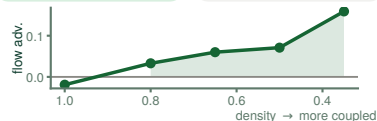
E When does it help? Coupling

coupled steps

flow > independent gate

marginalizable steps

tie (e.g. MetaQA 2-hop)



Measured: the advantage grows with coupling and crosses zero when steps are independent.

F Evaluate

- **constraint-density sweep** synthetic, 3 seeds
 - **MoE FFN layer** accuracy vs. compute
 - **MoE transformer** $E=6$ experts, $K=3$ layers
 - **MetaQA 2-hop KGQA** 43k entities, real data
- accuracy experts / layer illegal rate

Same encoder, same legal graph, same compute; only the router changes.